Forest Soils

Land Sciences

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Forest Soils (A06235)

Short Title: Forest Soils
Faculty Science and Computing
Department Land Sciences
Cluster Forestry
Author MSOTTOCORNOLA
Credits 5 **Level:** Introductory

Description of Module / Aims

This module is designed to equip students with a basic knowledge of soils, including soil formation, composition, structure and texture. The module will focus on soil-plant interaction, soil erosion, and the macronutrients and micronutrients involved in plant nutrition. Students will learn to identify a forest soil, carry out soil sampling, as well as a range of soil analyses, including pH and texture determination.

Programmes

FORS-0019	BSc in Forestry (WD_SFORS_D)	2 / 3 / M

Indicative Content

- Soil formation
- Soil identification and classification
- Soil ecology
- Soil texture and structure
- Plant nutrition
- Soil composition
- Soil sampling
- Soil-water interactions
- Soil-plant interaction

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the processes involved in the formation and classification of major soil types.
2. Describe soil physics, chemistry and biological properties and the uptake and role of macronutrients and micronutrients in the plants.
3. Explain soil composition, texture, structure and the environmental concerns connected to forest soils.
4. Identify the main forest soil types (peats, brown earths, gleys, podzols and lithosol) from soil pits and identify the most suitable tree species to be planted on different soils.
5. Complete surveys of forest soils according to the Forestry Schemes Manual procedure by the Forest Service.

Learning and Teaching Methods

- In-class lectures
- Site visits

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3
Continuous Assessment	50%	
Assignment	50%	4,5

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/ skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design and current knowledge limitations when carrying out assessments. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on assessments. Show initiative, individual thought, and enthusiasm in self-study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Field Trips	24	
Independent Learning	87	

Essential Material

- Ashman, M.R. and G. Puri. _Essential soil science_. Maiden, US: Blackwell Publishing, 2002.
- Brady, N.C. and R.R. Weil. _The Nature and Properties of Soils_. 14th ed.. London: Prentice-Hall, 2008.
- Environment Agency, . _Think soils manual_. UK: Environment Agency, 2007.
- Forest Service, . _Forestry Schemes Manual_. Johnstown Castle Estate, co. Wexford: Department of Agriculture, Food and the Marine, 2011.
- Foth, H.D. _Fundamentals of Soil Science_. 8th ed.. New York, US: John Wiley & Sons, 1990.
- Kennedy, F. _The identification of soils for forest management. Field_ _guide_. UK: Forestry Commission, 2002.

- Pritchett, W.L. and R.F. Fisher. _Properties and Management of forest soils_. 2nd ed.. New York, US: John Wiley & Sons, 1987.

Supplementary Material

- Pyatt, G. _An ecological site classification for forestry in Great Britain_. Wrecclesham, Farnham, Surrey, UK: Forestry Commission Research Division, 1995.
- Westerman, R.L., ed. _Soil Testing and Plant Analysis_. Madison, Wisconsin, US: Soil Science Society of America, 1990.